CODECHELLA MADRID 2026

LaLonde, Dehejia–Wahba, and Why the 2×2 Still Matters

Selection on observables, in the case that built the literature

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LESSON

1. LaLonde 1986

The National Supported Work Demonstration

A federally funded **randomized** job-training program.

1975–1979. Four target groups, including disadvantaged adult men.

Random assignment to treatment vs. control. Outcome: real earnings in 1978.

LaLonde's experimental benchmark

$$\widehat{\mathsf{ATT}}^{\mathsf{RCT}} \approx \$886$$

1978 dollars. Adult-male subsample.

Random assignment $\Rightarrow$ no selection. This is the truth.

Then LaLonde threw away the controls

He kept the treated — the 297 NSW men.

He replaced the controls with samples from the **CPS** and the **PSID**.

A simulation of what an applied researcher would face — no experiment, just observational data and methods.

Then he tried every method going at the time

- OLS with controls
- Two-step Heckman selection corrections
- Fixed effects on pre-program earnings
- Difference-in-differences

Could any of them recover $886?

No

Estimates ranged from $-$**$16,000** to $+$**$3,000**.

Most were negative. Most missed the benchmark by orders of magnitude.

LaLonde (1986), AER 76(4): 604–620.

The headline that shook applied econometrics

"The econometric procedures cannot replicate the experimental results."

This is the paper that made program evaluation move toward randomization and design.

LESSON

2. Dehejia & Wahba 2002

Dehejia & Wahba reopened the case

Rajeev Dehejia and **Sadek Wahba**,
Review of Economics and Statistics (2002).

They asked: does **propensity score matching** do better than LaLonde's methods?

First, they re-cut the experimental sample

They restricted to NSW men with earnings observed in **both 1974 and 1975**.

A smaller, slightly different population.

This is the subsample the homework uses: `lalonde_exp_panel.dta`.

A new experimental benchmark

$$\widehat{\mathsf{ATT}}^{\mathsf{RCT}}_{\mathsf{D\&W}} \approx \$1{,}794$$

Roughly **$1,700** in the way we usually report it.

Why bigger than LaLonde's $886? Different subsample, different ATT.

Why the headlines differ

- Both numbers come from the *same* NSW experiment.
- LaLonde used the full adult-male sample.
- D&W used a *subset* — only men with two years of pre-program earnings.
- Subsetting on a covariate changes the population whose ATT you're estimating.
- Same identification (random assignment). Different target group.

This is not a contradiction — it is the ATT defined on different units.

D&W's quasi-experimental result

With **propensity-score matching** on CPS / PSID controls...

they recovered $\approx \$1{,}700$.

A constructive answer to LaLonde: design plus matching can work.

LESSON

3. Why the ATT didn't move

A puzzle

We used **non-experimental controls** (CPS, PSID)
in the larger LaLonde–D&W literature.

And yet the ATT we target is still $\approx \$1{,}700$.

Why doesn't the ATT change when we change the controls?

Because the ATT is a property of the treated

$$\text{ATT} = \mathbb{E}[Y^1 - Y^0 \mid D = 1, \text{ Post}]$$

Only the $D = 1$ side appears.

The control units are just a *tool* to learn about $\mathbb{E}[Y^0 \mid D{=}1, \text{Post}]$.

Random assignment gave us a clean tool

Under random assignment:

$$\mathbb{E}[Y^0 \mid D{=}1] \;=\; \mathbb{E}[Y^0 \mid D{=}0]$$

So $\mathbb{E}[Y \mid D{=}0, \text{Post}]$ *is* the counterfactual we need.

No selection bias.

Non-experimental controls break that equality

CPS men and NSW participants are not interchangeable.

$$\mathbb{E}[Y^0 \mid D{=}1] \;\neq\; \mathbb{E}[Y^0 \mid D{=}0]$$

Selection bias enters.

NSW men had worse pre-program earnings. The CPS pool didn't.

But the *target* hasn't changed

$\mathbb{E}[Y^0 \mid D{=}1, \text{Post}]$ is a feature of the NSW men.

It does not depend on who we picked as controls.

What the controls control is *our estimate* of that target —
and the size of the bias.

So what is DiD for, in this story?

- In the RCT sample: DiD just *confirms* the post-period difference-in-means.
- In the non-experimental sample: DiD tries to *net out* the pre-existing level gap between NSW and CPS men.
- The truth is fixed at $1,700. The question is whether DiD's parallel-trends assumption is enough to recover it.

Did it?

The benchmark and the challenge

RCT benchmark: $\$1{,}794$

Naive DiD on the non-experimental panel: $\$3{,}621$

We can do better than additive controls. Let's go to the code.

LESSON

4. Now make it harder

Throw away the experimental controls

Keep the 297 NSW treated men.

Replace the controls with a random sample of American households from the **CPS**.

Data: lalonde_nonexp_panel.dta. Same treated rows. New controls.

Selection is severe

	1974	1975	1978
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15{,}992$)	\$14,017	\$13,651	\$14,847
Diff	$-\$11{,}921$	$-\$12{,}119$	$-\$8{,}498$

$$\mathbb{E}[Y^0 \mid D{=}1] \ll \mathbb{E}[Y^0 \mid D{=}0]$$

The level gap is huge — and so is the growth-rate gap. DiD doesn't fix that.

The 2×2: where \$3,621 comes from

	1974	1975 (pre)	1978 (post)
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15{,}992$)	\$14,017	\$13,651	\$14,847

NSW change: \$6,349 − \$1,532 = \$4,817
CPS change: \$14,847 − \$13,651 = \$1,196

DiD \$4,817 − \$1,196 = $\boxed{\$3{,}621}$

$$\$3{,}621 \neq \$1{,}794$$

NSW men bounce back fast from a deep dip. CPS men don't. DiD attributes the entire gap in growth rates to the program.

Naive DiD on this sample

$$\widehat{\delta}_{\text{naive DiD}} = +\$3{,}621$$

Right sign. *Wrong magnitude.* The benchmark is $\$1{,}794$.

NSW men had unusually low pre-earnings ($\$1{,}532$) and bounced back fast.
CPS controls didn't. DiD attributes the gap in growth rates entirely to treatment.

LESSON

5. Run the specs

Spec 0: Unconditional parallel trends, no covariates

Assumption. Naive parallel trends: $E[\Delta Y(0) \mid D{=}1] = E[\Delta Y(0) \mid D{=}0]$ unconditionally.

Run:

```
reg re i.post##i.ever_treated, robust
```

Compare to $1,794.

Spec A: Conditional PT, additive X

Assumption. Conditional parallel trends with one slope per covariate (pre = post).

Run:

```
global X "age␣agesq␣educ␣educsq␣marr␣nodegree␣black␣hisp␣re74␣re75␣u
reg re i.post##i.ever_treated $X, robust
```

Compare to $1,794.

Spec B: Conditional PT, Post $\times X$

Assumption. Conditional parallel trends, allowing X-effects to differ pre vs post. Use ## so the spec includes *both* the pre-period X effect and its post-period change.

Run:

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##c.re75 i.post##i.u74 i.post##i.u75, robust
```

Compare to $1,794. *Single-# forces pre-period X effect to zero — not what we want.*

Spec C: Saturated TWFE / OR (HIT 1997)

Assumption. Conditional parallel trends; impute $Y(0)$'s trend on controls only and average over treated.

Run — by hand:

```
reshape wide re, i(id) j(post)
gen dy = re1 - re0

reg dy $X if ever_treated == 0
predict dy_hat
gen delta_hat = dy - dy_hat if ever_treated == 1
summarize delta_hat if ever_treated == 1
```

Cross-check with `drdid`:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) reg
```

Compare to $1,794. *Two numbers should agree to the dollar.*

Spec D: IPW (Abadie 2005)

Assumption. Conditional PT; correct propensity score. Treated weighted by 1; controls weighted by $\hat{p}(X)/(1-\hat{p}(X))$.

Run — by hand:

```
logit ever_treated $X
predict phat, pr
quietly sum ever_treated
local pr_treat = r(mean)
gen w_ipw = (ever_treated - phat) / (1 - phat) / ‘pr_treat’
gen␣ipw_dy␣=␣w_ipw␣*␣dy
summarize␣ipw_dy
```

Cross-check with `drdid`:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

Compare to $1,794.

Spec E: DR-DiD (Sant'Anna & Zhao 2020)

Assumption. Conditional PT, AND *either* $\hat{\mu}_0$ or $\hat{p}(X)$ is correct (not both required).

Run — by hand (in wide-FD form, `phat/dy_hat` already computed):

```
gen dr_t = ever_treated * (dy - dy_hat) / ‘pr_treat’
gen␣dr_c␣=␣(1␣-␣ever_treated)␣*␣(phat/(1-phat))␣*␣(dy␣-␣dy_hat)␣/␣‘pr_treat’
quietly sum dr_t
local mt = r(mean)
quietly sum dr_c
display "DR␣ATT␣=␣" %9.0fc ‘mt’␣-␣r(mean)
```

Cross-check with `drdid`:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Compare to $1,794.

LESSON

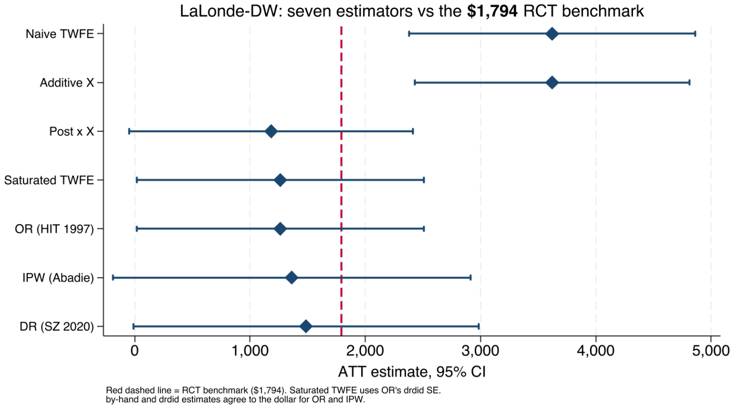
6. How close did each one get?

The comparison

Spec	ATT	Distance from $1,794
0. Naive TWFE (no X)	$+\$3{,}621$	$+\$1{,}827$
A. Additive X	$+\$3{,}621$	$+\$1{,}827$
B. Post$\times X$ (## syntax)	$+\$1{,}184$	$-\$610$
C. Saturated TWFE (FD)	$+\$1{,}263$	$-\$531$
OR by hand = `drdid reg`	$+\$1{,}263$	$-\$531$
D. IPW by hand = `drdid ipw`	$+\$1{,}361$	$-\$433$
E. DR by hand	$+\$1{,}496$	$-\$298$
DR `drdid dripw`	$+\$1{,}485$	$-\$309$
RCT benchmark	$+\$1{,}794$	—

Naive = Additive-X: the linear X control is absorbed. DR closest at $298 below. By-hand = `drdid` to the dollar for OR and IPW; DR within $11.

Seven specs, one benchmark



Naive and Additive-X CIs sit entirely above $1,794. Every design-aware spec contains it.

DR lands closest; the others bracket it

- **DR** ($1,496) sits **$298 below** — closest observational estimator.
- **IPW** ($1,361) sits $433 below.
- **OR** / **Saturated TWFE** ($1,263) sits $531 below — two paths, same number.
- **Post**$\times X$ ($1,184) sits $610 below — still misses Bias_{HTE}.
- **Additive** X ($3,621) doesn't move off Naive ($3,621) — the FE absorbed the linear X.

"Doubly robust" = robust to one misspecification, not best in every sample. DR's 298 below is selection on unobservables — the residual no observational X can close.

What this lab teaches you

- The **ATT is fixed** ($\approx +\$1{,}794$). The controls only change the bias.
- **Additive X barely helps:** \$3,621 vs naive \$3,621. You need a design-aware estimator.
- Post$\times X$, OR, IPW, DR offer four different uses of the same X, all closer to truth.
- On this dataset, **DR-DiD lands closest** at \$298 below \$1,794.
- No method is magic. Each rests on an assumption you can name.

What additive X actually assumed

The additive-X spec $Y_{it} = \alpha + \delta(D_i \cdot \text{post}_t) + X_i' \beta + \varepsilon$ was a **TWFE specification choice**. Two restrictions came along free of charge:

1. **Homogeneous treatment effect in X.** One scalar δ. No room for $\delta(X)$.
2. **No time-varying X effect.** One β, used in both pre and post. If X's relationship with $Y(0)$ shifts between periods, the spec can't see it.

Neither restriction is identification. Both are specification.

Two TWFE relaxations

We saw both in the synthetic simulation. Both are still TWFE — just less restrictive parameterizations.

Post $\times X$. Let β differ pre vs post:

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \varepsilon$$

Relaxes restriction (2). Still imposes (1): one scalar δ.

Fully saturated. Let β differ pre/post *and* let δ depend on X:

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \beta_3 XD + \beta_4 DTX + \varepsilon$$

Relaxes both. The ATT is now $\delta + \beta_4\,\mathbb{E}[X \mid D{=}1]$ — average the X-specific effect over the treated.

The punchline

Additive X failed on LaLonde

not because the data are weird,

but because the **TWFE specification** was too restrictive.

Fully saturated TWFE = the regression form of HIT 1997 outcome regression.

Same estimator. Same ATT. Different machinery.

Spec C (FD with $D \times X$ interactions) and OR-by-hand returned the same dollar: $1,263.

The ATT lives on the **treated.**

The controls only tell you

whether your method
can **see it.**